Learning Objective 8.3: I know the forms of Maxwell's equations for time-varying fields and use them to calculate electric and magnetic fields for a variety of physical systems.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A pickup coil.** A square loop 20 cm on a side lies in the xy plane in a uniform field $\mathbf{B} = \hat{\mathbf{z}}\, 0.5 \sin(120\pi t)$ mT.

   (a) Find the amplitude of the voltage induced around the loop.

   The flux is $\Phi = AB = (0.2)^2(0.5 \times 10^{-3}) \sin(120\pi t)$ Wb. Faraday's law gives

   $$\oint_C \mathbf{E}\cdot d\mathbf{l} = -\frac{d\Phi}{dt}$$

   $$= -(0.04)(0.5 \times 10^{-3})(120\pi)\cos(120\pi t)$$

   $$= -(7.54\ \text{mV})\cos(120\pi t)$$

   amplitude = 7.54 mV

   (b) We replace the loop with a coil of 50 turns of the same size. Find the new amplitude.

   Each turn links the same flux, and the turns are in series:

   $$50 \times 7.54\ \text{mV} = 377\ \text{mV}$$

   amplitude = 377 mV

   (c) Just after $t = 0$, B is positive and increasing. Seen from $+z$, which way does the induced current flow in the single loop? Explain with Lenz's law.

   Clockwise. Just after $t = 0$ the cosine is positive, so the circulation counterclockwise is negative and the current flows clockwise. A clockwise current makes a field along $-\hat{\mathbf{z}}$ inside the loop, which opposes the growing flux along $+\hat{\mathbf{z}}$.

2. **Checking a pair of fields.** A lossless, source-free region has $\epsilon_r = 4$ and $\mu_r = 1$. A proposed electric field is $\mathbf{E} = \hat{\mathbf{y}}\, 10\cos(2\pi \times 10^9\, t - \beta z)$ V/m.

(a) Use Faraday's law to find the magnetic field **H** that must accompany **E**, in terms of β.

E_y depends only on z, so only the $\hat{\mathbf{x}}$ component of the curl survives:

$$\nabla \times \mathbf{E} = -\hat{\mathbf{x}}\,\frac{\partial E_y}{\partial z} = -\hat{\mathbf{x}}\, 10\beta \sin(\omega t - \beta z) = -\mu\frac{\partial \mathbf{H}}{\partial t}$$

$$\mathbf{H} = -\hat{\mathbf{x}}\,\frac{10\beta}{\mu\omega}\cos(\omega t - \beta z)$$

(b) Use the Ampere–Maxwell law to find the value of β for which the pair satisfies Maxwell's equations, and the wavelength.

β = 41.9 rad/m

λ = 15.0 cm

$$\nabla \times \mathbf{H} = \hat{\mathbf{y}}\,\frac{\partial H_x}{\partial z} = -\hat{\mathbf{y}}\,\frac{10\beta^2}{\mu\omega}\sin(\omega t - \beta z)$$

$$\epsilon\frac{\partial \mathbf{E}}{\partial t} = -\hat{\mathbf{y}}\, 10\epsilon\omega \sin(\omega t - \beta z)$$

Matching gives $\beta = \omega\sqrt{\mu\epsilon} = \omega\sqrt{\epsilon_r}/c$:

$$\beta = \frac{(2\pi \times 10^9)(2)}{3 \times 10^8} = 41.9 \text{ rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = 15.0 \text{ cm}$$

(c) Find the amplitude of **H** and its direction.

H_0 = 53.1 mA/m

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{377}{\sqrt{4}} = 188\,\Omega$$

$$H_0 = \frac{10}{188} = 53.1 \text{ mA/m}, \quad \text{along } -\hat{\mathbf{x}}$$

(d) Show that the pair also satisfies both of Gauss's laws.

D_y depends only on z and t, so $\nabla\cdot\mathbf{D} = \partial D_y/\partial y = 0 = \rho_v$. B_x depends only on z and t, so $\nabla\cdot\mathbf{B} = \partial B_x/\partial x = 0$.

3. **Conservation of charge.** In a region, the current density is $\mathbf{J} = \hat{\mathbf{r}}\, kr$ A/m^2 with $k = 2$ A/m^3, in spherical coordinates.

   (a) Find the rate of change of the charge density, $\partial\rho_v/\partial t$.

   The continuity equation, with the spherical divergence from Lesson 17:

   $$\frac{\partial \rho_v}{\partial t} = -\nabla\cdot\mathbf{J} = -\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\, kr\right) = -3k = -6 \text{ C/(m}^3\text{s)}$$

   $\partial\rho_v/\partial t$ = -6 C/(m^{3}s)

   (b) Find the current out of a sphere of radius 10 cm, and show that it equals the rate at which the charge inside decreases.

   current = 25.1 mA

   $$\oint \mathbf{J}\cdot d\mathbf{s} = (kr)(4\pi r^2) = 4\pi k r^3$$

   $$= 4\pi(2)(0.1)^3 = 25.1 \text{ mA}$$

   $$-\frac{dQ}{dt} = \frac{4}{3}\pi r^3 (3k) = 4\pi k r^3 = 25.1 \text{ mA}$$

   (c) Explain why Ampere's law without the displacement current could not describe this region.

   The divergence of a curl is zero, so $\nabla \times \mathbf{H} = \mathbf{J}$ would force $\nabla\cdot\mathbf{J} = 0$. Here $\nabla\cdot\mathbf{J} = 6$ A/m^3, because charge is leaving the region. The displacement current $\partial\mathbf{D}/\partial t$ supplies the term whose divergence is $\partial\rho_v/\partial t$ and restores the balance.

4. **An ideal transformer.** A transformer runs from a 240 V rms, 50 Hz line with $N_1 = 960$ turns on the primary and supplies 24 V rms to an $8\ \Omega$ load. The core has a cross-sectional area of 10 cm^2.

   (a) Find the number of secondary turns and both currents.

$$N_2 = N_1 \frac{V_2}{V_1} = 960 \times \frac{24}{240} = 96$$

$$I_2 = \frac{24}{8} = 3\ \text{A}, \qquad I_1 = I_2 \frac{N_2}{N_1} = 0.3\ \text{A}$$

N_2 = 96

I_1 = 0.3 A

   (b) Find the peak flux in the core and the peak flux density.

With $v_1 = N_1\, d\Phi/dt$ and $V_{1,\text{pk}} = 240\sqrt{2} = 339$ V:

$$\Phi_{\text{pk}} = \frac{V_{1,\text{pk}}}{N_1 \omega} = \frac{339}{(960)(2\pi \times 50)} = 1.13\ \text{mWb}$$

$$B_{\text{pk}} = \frac{\Phi_{\text{pk}}}{A} = \frac{1.13 \times 10^{-3}}{10 \times 10^{-4}} = 1.13\ \text{T}$$

B_{pk} = 1.13 T

   (c) A student proposes saving copper by using $N_1 = 240$ and $N_2 = 24$, the same ratio. Explain what happens to the core.

The peak flux density scales as $1/N_1$, so it rises by a factor of 4, to 4.50 T. That is far above the 1.5 to 1.7 T at which a silicon-steel core saturates, so μ collapses, the magnetizing current grows sharply, and the ideal model no longer holds.

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